

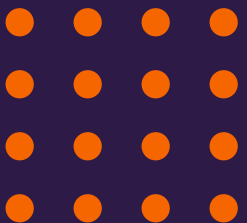


# Model Selection and Cross-Validation

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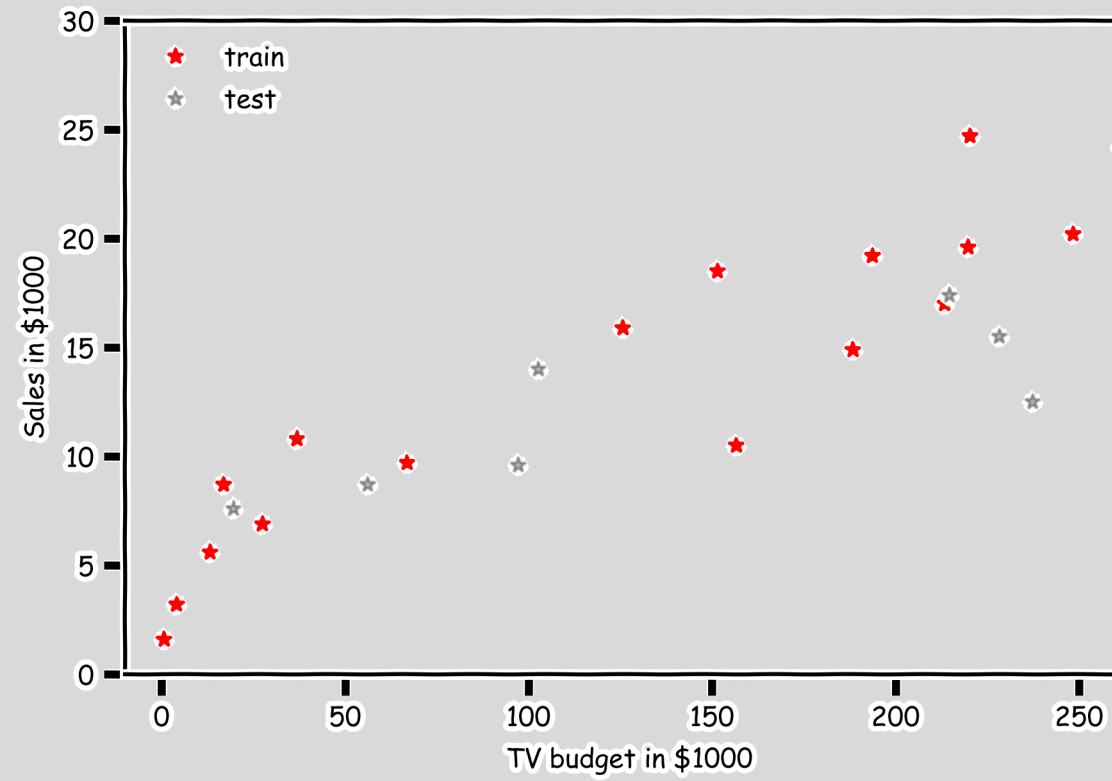


# Lecture Outline

- **Overfitting**
- **Model Selection**
- **Cross Validation**
- **Bias vs. Variance**

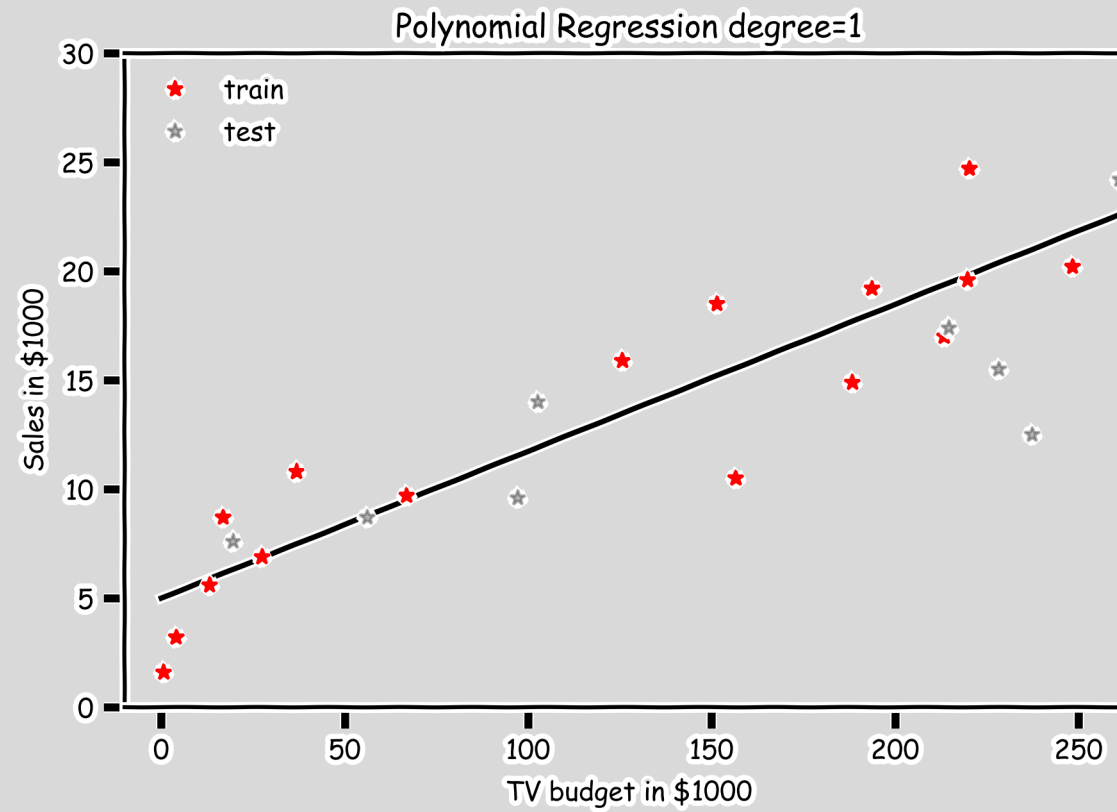


# Overfitting



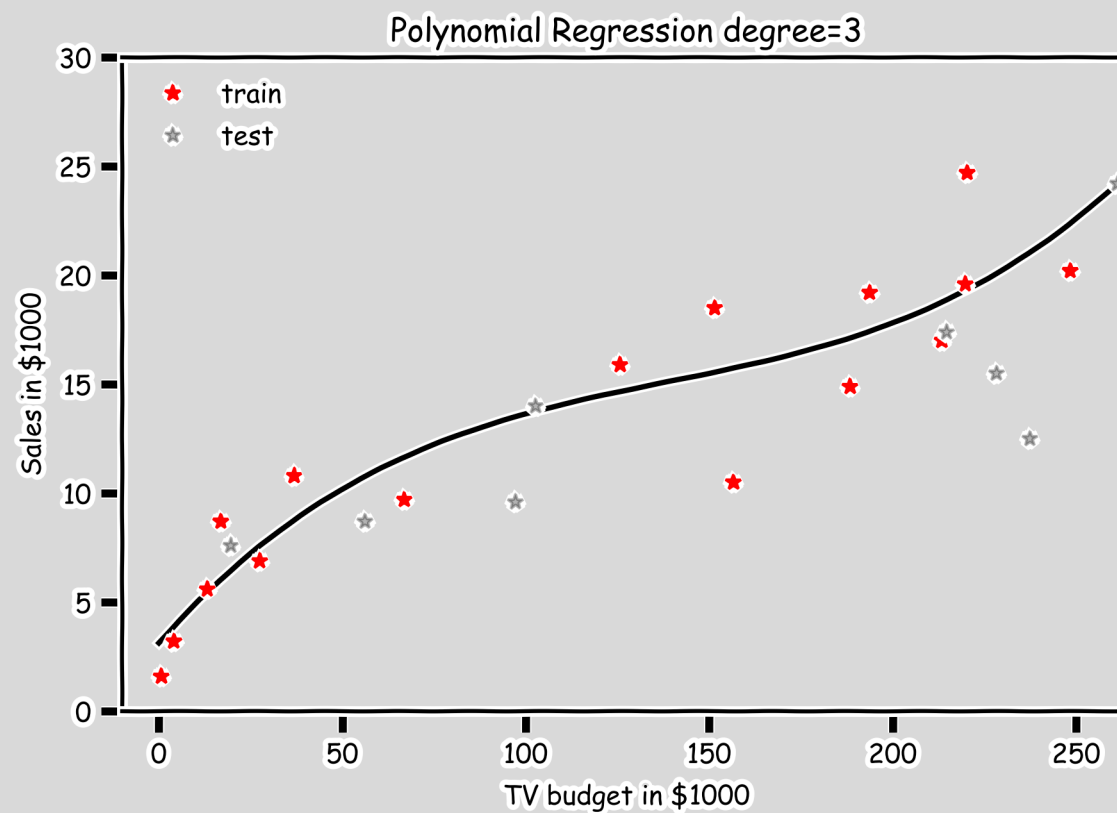


# Overfitting



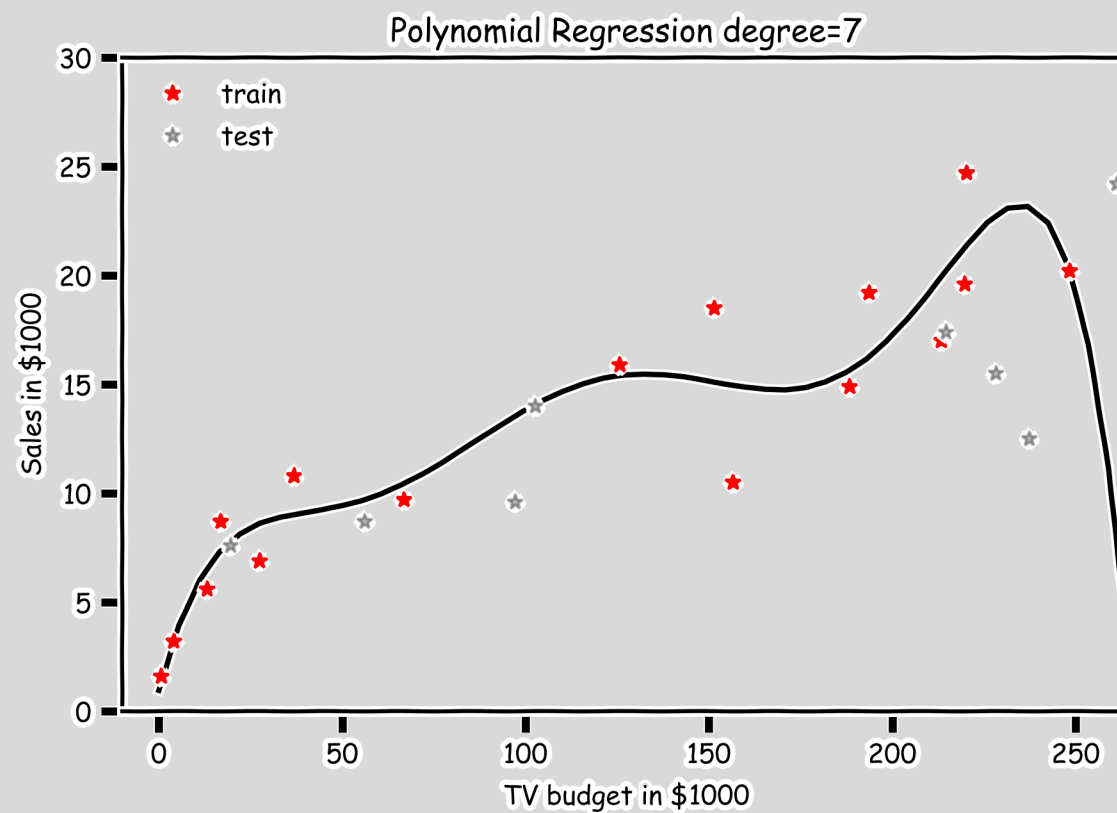


# Overfitting



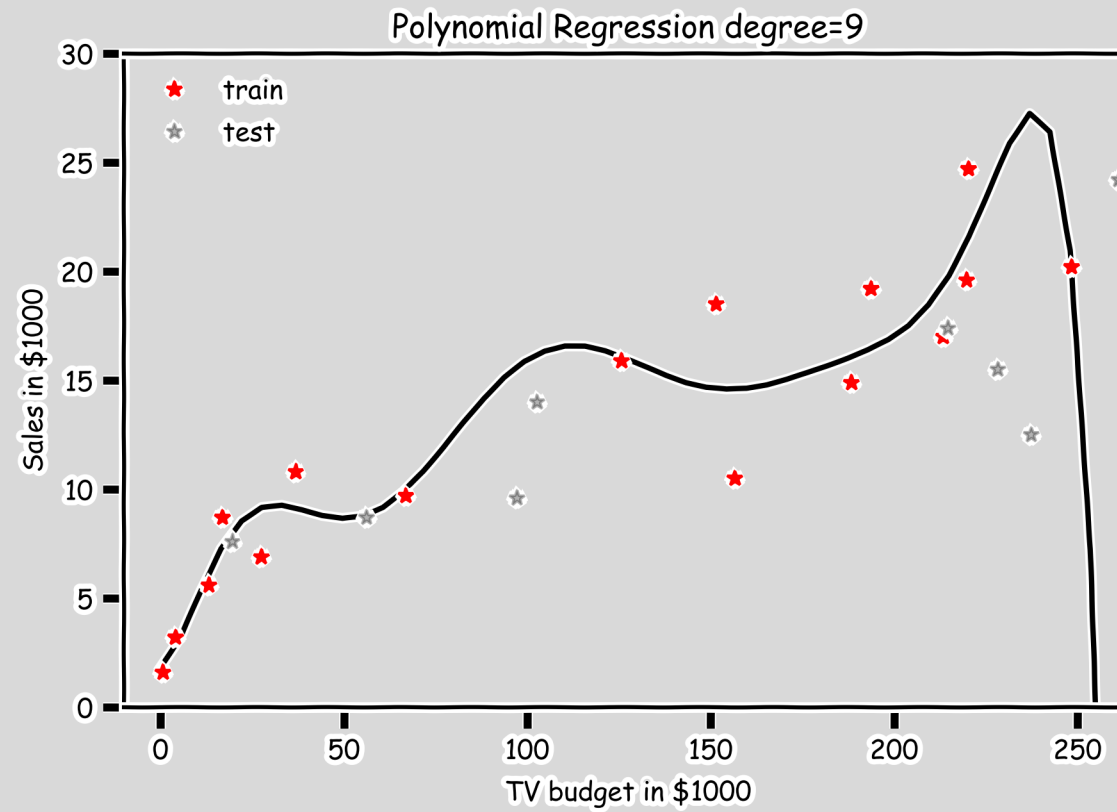


# Overfitting



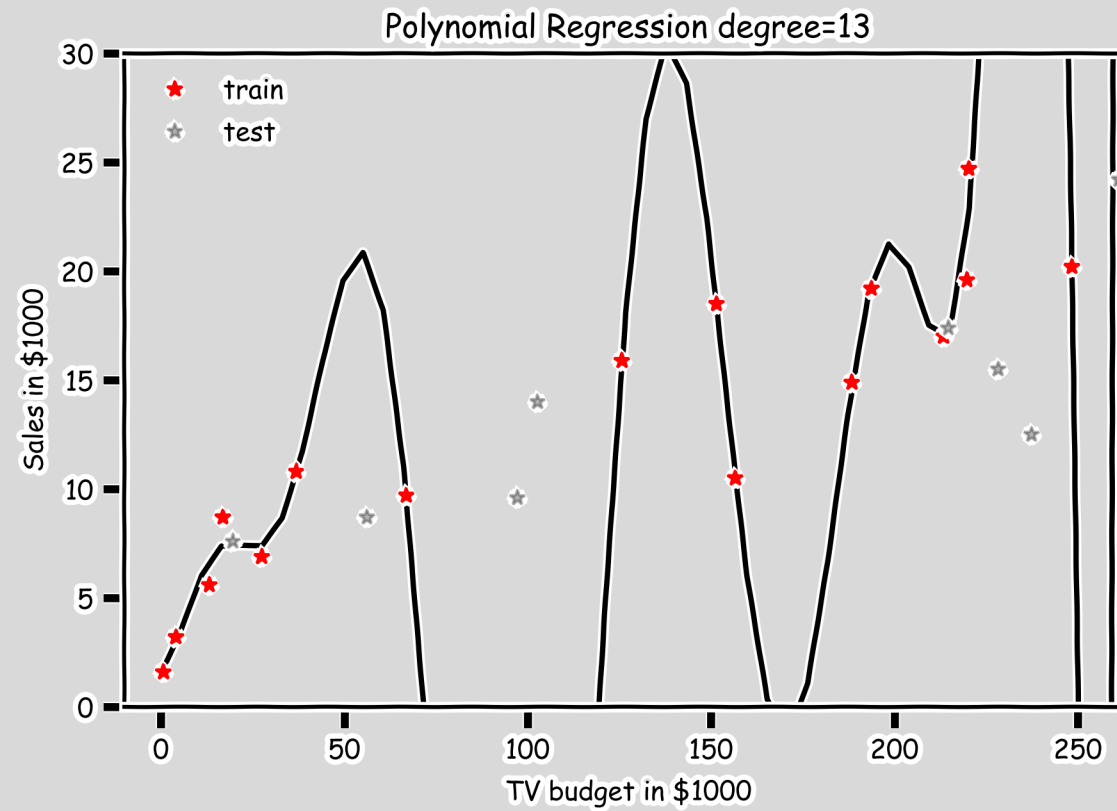


# Overfitting



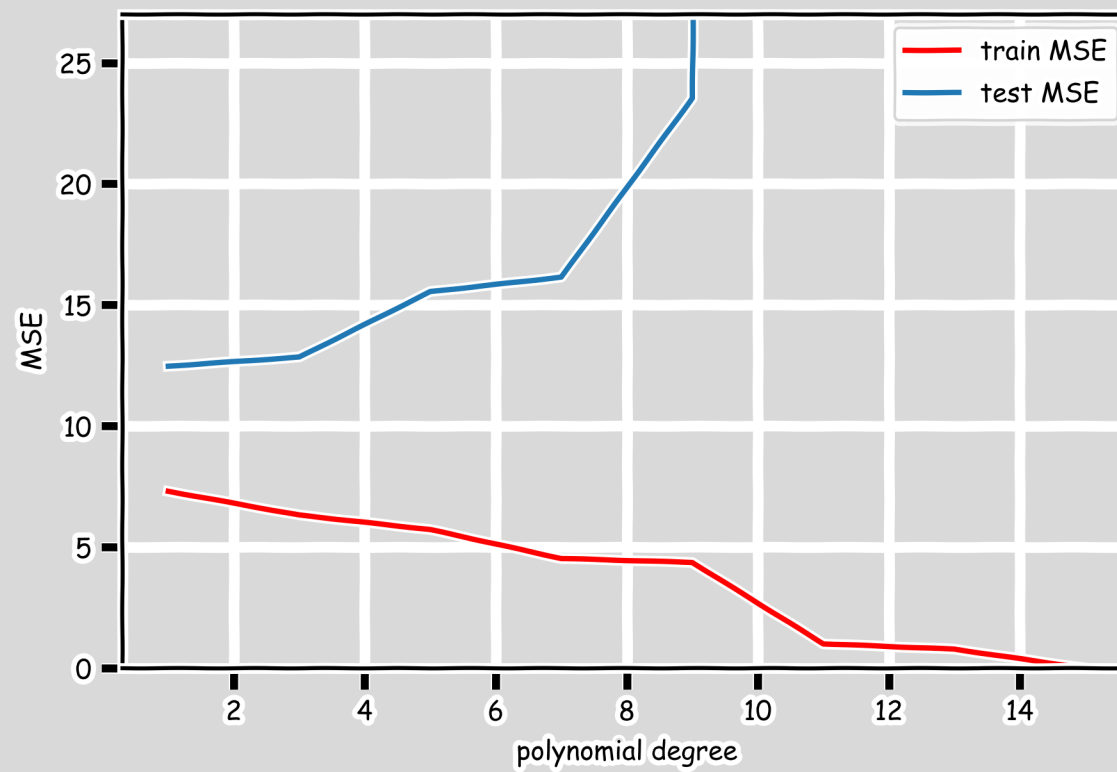


# Overfitting





# Validation



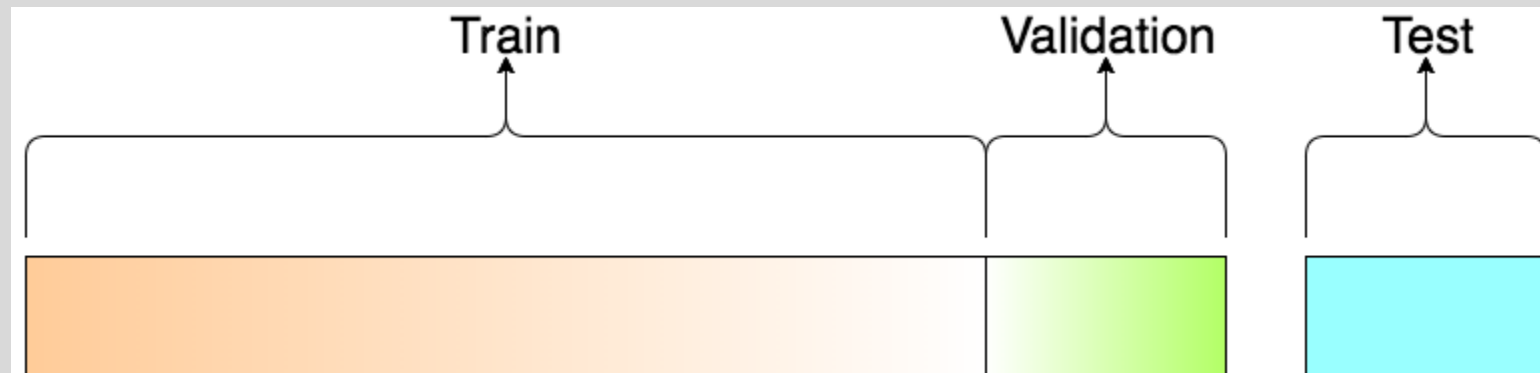
# Train-Validation-Test

## Question:

How would you report the performance of the model?

$R^2_{\text{test}} = 0.52$

$R^2_{\text{train}}(\text{degree}=1) = 0.83$





# Lecture Outline

- Overfitting
- **Model Selection**
- Cross Validation
- Bias vs. Variance

# Model Selection

**Model selection** is the application of a principled method to determine the complexity of the model, e.g., choosing a subset of predictors, choosing the degree of the polynomial model etc.

A strong motivation for performing model selection is to avoid overfitting, which we saw can happen when:

- there are too many predictors:
  - the feature space has high dimensionality
  - the polynomial degree is too high
  - too many cross terms are considered
- the coefficients values are too **extreme (we have not seen this yet)**



# Model Selection

## Question:

How many different models when considering  $J$  predictors?



# Model Selection

Example: 3 predictors ( $X_1, X_2, X_3$ )

Models with 0 predictor:

- M0:

Models with 1 predictor:

- M1:  $X_1$
- M2:  $X_2$
- M3:  $X_3$

Models with 2 predictors:

- M4:  $\{X_1, X_2\}$
- M5:  $\{X_2, X_3\}$
- M6:  $\{X_3, X_1\}$

Models with 3 predictors:

M7:  $\{X_1, X_2, X_3\}$

$2^J$  Models



# Stepwise Variable Selection & Cross Validation

Selecting optimal subsets of predictors (including choosing the degree of polynomial models) through:

- stepwise variable selection - iteratively building an optimal subset of predictors by optimizing a fixed model evaluation metric each time,
- validation - selecting an optimal model by evaluating each model on validation set.

We will also address the issue of discouraging extreme values in model parameters later.

# Stepwise Variable Selection: Forward Method

In **forward selection**, we find an ‘optimal’ set of predictors by iterative building up our set.

1. Start with the empty set  $P_0$ , construct the null model  $M_0$ .
2. For  $k = 1, \dots, J$ :
  - 2.1 Let  $M_{k-1}$  be the model constructed from the best set of  $k - 1$  predictors,  $P_{k-1}$ .
  - 2.2 Select the predictor  $X_{n_k}$ , not in  $P_{k-1}$ , so that the model constructed from  $P_k = X_{n_k} \cup P_{k-1}$  optimizes a fixed metric (this can be p-value, F-stat; validation MSE,  $R^2$ , or AIC/BIC on training set).
  - 2.3 Let  $M_k$  denote the model constructed from the optimal  $P_k$ .
3. Select the model  $M$  amongst  $\{M_0, M_1, \dots, M_J\}$  that optimizes a fixed metric (this can be validation MSE,  $R^2$ ; or AIC/BIS on training set)



# Stepwise Variable Selection: Computational Complexity

How many models did we evaluate?

- 1st step,  $J$  Models
- 2nd step,  $J-1$  Models (add 1 predictor out of  $J-1$  possible)
- 3rd step,  $J-2$  Models (add 1 predictor out of  $J-2$  possible)
- ...

$$O(J^2) \ll 2^J \text{ for large } J$$